

TIMOTHY METZGER

+1(314) 668-0558 ♦ Eureka, MO

tmetzger8@gmail.com ♦ <https://timmetzger.github.io/>

EDUCATION

M.S. Computer Science, Georgia Institute of Technology 2022-2024
Specialization: Interactive Intelligence GPA: 3.90

B.S. Mechanical Engineering, Saint Louis University 2019 - 2022
Minor in Engineering Mathematics GPA: 4.0
Summa Cum Laude

SKILLS

Technical Skills	Python, C++, C#, AWS, Ada, SQL, Jira, Confluence, Git, MATLAB, MS Office
Libraries	NumPy, SciPy, Pandas, Matplotlib, PyTorch, VectorCast, Boost, Catch2
Soft Skills	Problem Solving, Coordination, Communication, Teamwork, Management
Certifications	Professionally Licensed EIT - MO

PROFESSIONAL EXPERIENCE

Senior Real-Time Software Engineer July 2025 - Present
LSEG Data & Analytics *St. Louis, MO*

- Independently developed real-time data transform and normalization software in C++ for financial data
- Deployed software to cloud environment using AWS resources including EC2, SQS, S3, and ElastiCache
- Developed AI agent pipelines to accelerate software development throughout the software development life cycle
- Led efforts to onboard existing engineers to accelerated development using AI agents
- Organized code reviews using GitLab integrated with SonarQube to ensure maintainable and secure software

Real-Time Software Engineer - F22 May 2024 - July 2025
Boeing *St. Louis, MO*

- Implemented new capabilities into legacy systems containing 1M+ lines of code in Ada and C++
- Led in person code reviews using BitBucket to ensure software met customer requirements
- Performed system validation and verification through extensive testing procedures using C#
- Orchestrated the successful delivery of monthly software releases involving multiple companies
- Developed new algorithms utilizing electronic warfare sensors for enemy threat identification
- Administered knowledge transfer sessions to improve team efficiency and effectiveness
- Created comprehensive documentation to verify complex system behavior using Confluence

Algorithmic Stock Trader Jan 2019 - May 2024
Personal *Eureka, MO*

- Utilized SciKitLearn and Pytorch to develop machine learning based trading strategies
- Created multiple algorithmic trading strategies combining custom indicators with reinforcement learning

PROJECTS

N-Directional A* Search of Atlanta. Implemented a version of N-Directional A* search as a potential solution to the infamous Traveling Salesman Problem. The algorithm makes use of optimization techniques commonly used in bi-directional search, such as landmarking and shortcuts, to provide an optimal solution connecting N points.